

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

***CO1:** Apply object-oriented concepts and software development life cycle models to design structured and modular software systems. (BL2, BL3)*

Assessment Tool Weightage: 60%

Dr Ramamoorthy

QUESTIONS: 12

Total Marks:60

Annexure A

CONCEPT MAPPING

Code of Conduct:

I (K.Deepak / 192320078) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

CONCEPT MAPPING

1. Construct a Concept Map

Given:

Requirement Analysis → _____ → Coding → Testing → _____ → Maintenance

Answer:

The missing concepts are:

- Design
- Deployment

Explanation:

After gathering the requirements, the system is designed before coding begins. Once testing is completed successfully, the software is deployed so that users can start using it.

Analysis:

This sequence follows the Software Development Life Cycle (SDLC). Each phase depends on the previous one, making software development organized, efficient, and easier to maintain.

2. Construct a Concept Map

Given:

Object-Oriented Principles

→ Abstraction

→ _____

→ Inheritance

→ _____

Answer:

The missing concepts are:

- Encapsulation
- Polymorphism

Explanation:

Encapsulation protects data by combining it with methods inside a class. Polymorphism allows the same method to perform different tasks based on the object.

Analysis:

These principles improve security, code reuse, flexibility, and make software easier to develop and maintain.

3. Construct a Concept Map

Given:

Class → Blueprint

Object → Instance

Method → _____

Attribute → _____

Answer:

The missing concepts are:

- Behavior
- State

Explanation:

Methods define what an object can do, while attributes store the data that describe the object.

Analysis:

These concepts help represent real-world objects accurately and improve software organization.

4. A Concept Map Shows

Given:

Software Quality

→ Reliability

→ _____

→ Maintainability

Answer:

The missing concept is:

- Usability

Explanation:

Usability refers to how easily users can understand and operate the software. Reliable and maintainable software provides a better user experience.

Analysis:

Good software should be reliable, user-friendly, and easy to update whenever required.

5. UML Concept Map

Given:

UML

→ Structural Diagrams → _____

→ Behavioral Diagrams → Activity Diagram

→ Interaction Diagrams → _____

Answer:

The missing concepts are:

- Object Diagram
- Communication Diagram

Explanation:

Object diagrams represent objects at a specific time, while communication diagrams show how objects exchange messages.

Analysis:

These UML diagrams improve system visualization and simplify software design.

6. Given:

Requirements ✓ → Design ✓ → Coding ✗ → Testing ✗ → Deployment ✗

Explanation:

The failure occurs during the Coding phase because implementation errors are introduced.

Analysis:

Coding mistakes affect testing and deployment, resulting in poor software quality and delays.

Solution:

- Identify and correct coding errors before moving to the testing phase.
- Follow coding standards and best programming practices.
- Conduct code reviews and debugging regularly.
- Perform unit testing to verify that each module works correctly.
- Re-test the software after fixing errors to ensure it is ready for deployment.

7. Construct a Concept Map

Given:

Problem Identification → _____ → Design → Implementation → _____

Answer:

The missing concepts are:

- Requirement Analysis
- Testing

Explanation:

Requirement analysis identifies user needs, while testing ensures the developed software works correctly.

Analysis:

These stages reduce development errors and improve software quality.

8. Construct a Concept Map

Given:

Object-Oriented Features

→ Class

→ _____

→ Message Passing

→ _____

Answer:

The missing concepts are:

- Object
- Dynamic Binding

Explanation:

Objects are created from classes. Dynamic binding allows the method to be selected during program execution.

Analysis:

These features increase flexibility, support runtime polymorphism, and improve maintainability.

9. Construct a Concept Map

Given:

Software Development Models

→ Waterfall

→ _____

→ Spiral

→ _____

Answer:

The missing concepts are:

- Incremental Model
- Agile Model

Explanation:

Incremental development delivers software in small parts, while Agile focuses on continuous customer feedback and frequent improvements.

Analysis:

Both models reduce risks and help deliver high-quality software.

10. Construct a Concept Map

Given:

Class Diagram

→ Class

→ _____

→ Association

→ _____

Answer:

The missing concepts are:

- Attribute
- Method

Explanation:

Attributes store the properties of a class, while methods define the actions performed by the class.

Analysis:

Class diagrams clearly represent the structure and functionality of an object-oriented system.

11. Construct a Concept Map

Given:

Software Engineering Principles

→ Modularity

→ _____

→ Reusability

→ _____

Answer:

The missing concepts are:

- Maintainability
- Scalability

Explanation:

Maintainability makes software easier to update, while scalability allows the system to handle increased users or workload.

Analysis:

These principles improve software performance and support future growth.

12. Given:

Requirement Analysis ✓ → System Design ✓ → Coding ✓ → Integration ✗ →
Deployment ✗

Explanation:

The failure occurs during the Integration phase because individual modules do not work correctly together.

Analysis:

Integration problems delay deployment, increase debugging effort, and reduce overall system performance.

Solution:

- Perform integration testing to identify communication issues between modules.
- Verify that all module interfaces are compatible and functioning correctly.
- Fix integration and compatibility errors before deployment.
- Test the complete system to ensure all modules work together smoothly.
- Deploy the software only after successful integration testing.

CONCEPT MAPPING

RUBRICS FOR CALCULATION

CONCEPT MAPPING

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand